$$\frac{d}{dt}\begin{pmatrix} f_3 \\ f_7 \\ e_{11} \\ f_{13} \end{pmatrix} = \begin{pmatrix} -\frac{\mathrm{Re}\, l}{L} & -\frac{K}{L} & 0 & 0 \\ \frac{K}{J} & -\frac{R_{bearing}}{J} & -\frac{D}{2} & 0 \\ -\frac{D}{2C_{elasticity}} & -\frac{1}{C_{elasticity}} & 0 & 0 \\ 0 & 0 & \frac{1}{m} & 0 \end{pmatrix} \begin{pmatrix} f_3 \\ f_7 \\ e_{11} \\ f_{13} \end{pmatrix} + \begin{pmatrix} \frac{1}{L} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -\frac{1}{m} \end{pmatrix} \begin{pmatrix} u_{source} \\ mg \end{pmatrix}$$